

An Effective Two-Stage Column Generation Strategy for Solving Unit Commitment

Zhao-Gang Su

State Key Laboratory of Scientific and Engineering Computing
Academy of Mathematics and Systems Science, CAS

Joint work with **Sheng-Jie Chen, Yu-Fei Li, Liang Chen, Yu-Hong Dai**

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Agenda

- ① Introduction
- ② Extended Formulation of UC
- ③ A Two-stage Column Generation Strategy for UC
- ④ Numerical Experiment
- ⑤ Summary and Future Work

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Unit Commitment

- ▶ Schedule a set of electricity generators in a given period, with

- ① Generation limits
- ② Ramping constraints
- ③ Minimum up / down time
- ④ Spinning reserve

to minimize total production cost

- ▶ Application: **Day-ahead** market clearing in electricity market
- ▶ Importance of studying unit commitment
 - Solving approach change: Lagrangian Relaxation (-2006) → **Mixed Integer Linear Programming** (2006-)
 - Saving **\$5 billion** annually in U.S.A. [O'Neill, 2017]

Unit Commitment Problem

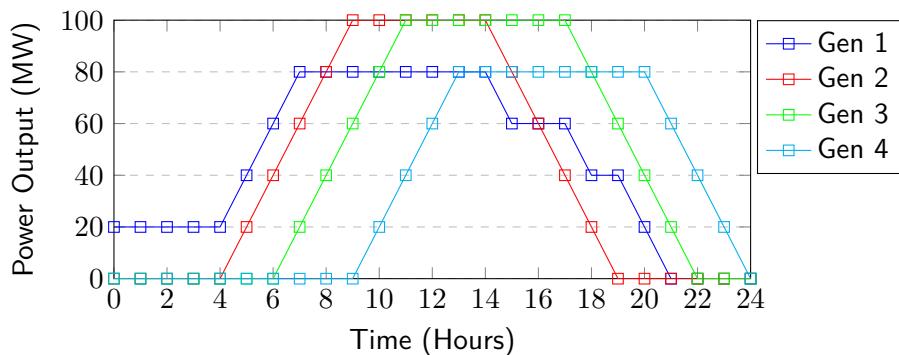


Figure: Example for 4-unit schedule in 24 hours

Unit Commitment Problem: 3-bin Formulation

► Variables

- p_t^i : Power output of unit i at time t , $t \in [1, T], i \in \mathcal{G}$
 - x_t^i : State 0-1 variable, $t \in [1, T], i \in \mathcal{G}$
 - s_t^i : Turn-on 0-1 variable, $t \in [2, T], i \in \mathcal{G}$
 - z_t^i : Turn-off 0-1 variable, $t \in [2, T], i \in \mathcal{G}$
- $$[a, b] = \{a, a + 1, \dots, b\}, a \leq b, a, b \in \mathbb{Z}_+$$

► Constants

- F_t^i : Production cost of unit i at time t
- C_t^i : Start-up cost of unit i at time period t
- $\overline{P}^i / \underline{P}^i$: Maximum/minimum power output of unit i
- $\bar{u}^i / \bar{v}^i$: Maximum start-up/shut-down production of unit i
- δ^i / η^i : Ramp up/down limit of unit i
- T_{on}^i / T_{off}^i : Minimum up/down time of unit i
- D_t : System load demand in time period t
- R_t : Spinning reserve demand in time period t

Unit Commitment Problem: 3-bin Formulation

$$\min \sum_{t=1}^T \sum_{i \in \mathcal{G}} (F_t^i p_t^i + C_t^i s_t^i) \quad (1)$$

s.t.

$$\begin{cases} p_t^i \geq \underline{P}^i x_t^i, & t \in [1, T], i \in \mathcal{G}, \\ p_t^i \leq \overline{P}^i x_t^i - (\overline{P}^i - \underline{u}^i) s_t^i - [\underline{u}^i - \overline{v}^i]^+ z_{t+1}^i, & t \in [2, T], i \in \mathcal{G} \\ p_{t-1}^i \leq \overline{P}^i x_{t-1}^i - (\overline{P}^i - \overline{v}^i) z_t^i - [\overline{v}^i - \underline{u}^i]^+ s_t^i, & t \in [2, T], i \in \mathcal{G} \end{cases} \quad (2)$$

$$\begin{cases} p_t^i - p_{t-1}^i \leq (\underline{u}^i - \underline{P}^i - \delta^i) s_t^i \\ \quad + (\underline{P}^i + \delta^i) x_t^i - \underline{P}^i x_{t-1}^i, & t \in [2, T], i \in \mathcal{G} \\ p_{t-1}^i - p_t^i \leq (\overline{v}^i - \underline{P}^i - \eta^i) z_t^i \\ \quad + (\underline{P}^i + \eta^i) x_{t-1}^i - \underline{P}^i x_t^i, & t \in [2, T], i \in \mathcal{G} \end{cases} \quad (3)$$

$$p_t^i \in \mathbb{R}_+, x_t^i \in \{0, 1\}, s_t^i \in \{0, 1\}, z_t^i \in \{0, 1\} \quad t \in [1, T], i \in \mathcal{G} \quad (6)$$

$$\begin{cases} x_t^i - x_{t-1}^i = s_t^i - z_t^i, & t \in [2, T], i \in \mathcal{G} \\ \sum_{j=t-T_{on}^i+1}^t s_j^i \leq x_t^i, & t \in [T_{on}^i + 1, T], i \in \mathcal{G} \\ \sum_{j=t-T_{off}^i+1}^t z_j^i \leq 1 - x_t^i, & t \in [T_{off}^i + 1, T], i \in \mathcal{G} \end{cases} \quad (4)$$

$$\begin{cases} \sum_{i \in \mathcal{G}} p_t^i \geq D_t, & t \in [1, T] \\ \sum_{i \in \mathcal{G}} \overline{P}^i x_t^i \geq D_t + R_t, & t \in [1, T]. \end{cases} \quad (5)$$

- ▶ Polyhedral study
 - Proved that the minimum up/down formulation is tightest [[Rajan et al., 2005](#)]
 - LP description of 2-period ramp-up polytope [[Damci-Kurt et al., 2016](#)]
 - Convex hull description for single unit based on dynamic programming [[Bacci et al., 2024](#)]
- ▶ Tightening formulation
 - Strengthened ramping inequalities [[Ostrowski et al., 2011](#)]
 - Strengthened generation limits inequalities [[Morales-España et al., 2013](#)]
 - Time-window formulation for multi-period UC [[Yang et al., 2022](#)]
 - Review of various UC formulations [[Knueven et al., 2020](#)]
- ▶ Surrogate Lagrangian relaxation
 - Development of SLR [[Bragin et al., 2015](#)]
 - SLR for large UC with special units [[Sun et al., 2018](#)]
 - Quantum SLR for UC [[Feng et al., 2022](#)]

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Structure of 3-bin Formulation

- ▶ Constraints (2)-(4) are **generator-wise**, i.e. they couple only the variables of each generator across time periods
- ▶ Constraints (5) link the variables of different generators within the same time period

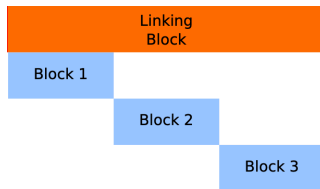


Figure: Structure of 3-bin formulation

Linking constraints (5) obstructs the exploitation of **parallel computing** capabilities.

Schedule of Units

For each unit $i \in \mathcal{G}$,

- ▶ $w^i \in \{0, 1\}^T$: **schedule** of unit i representing the on/off states of unit i over T periods
- ▶ $W(i)$: set of all possible schedules of unit i (schedules must satisfy constraints (4))

For each schedule w of unit $i \in \mathcal{G}$, define $\theta_w^i \in \{0, 1\}$ as **indicator variable**:

- ▶ if schedule w is selected from $W(i)$: $\theta_w^i = 1$
- ▶ if schedule w is not selected from $W(i)$: $\theta_w^i = 0$

By using variables θ_w^i and p_t^i while discarding variables x_t^i, s_t^i, z_t^i , an **extended formulation** of UC is obtained.

Extended Formulation of UC

$$\min \sum_{t=1}^T \sum_{i \in \mathcal{G}} F_t^i p_t^i + \sum_{i \in \mathcal{G}} \sum_{w \in W(i)} C_w^i \theta_w^i \quad (7)$$

s.t.

$$\begin{cases} \sum_{w \in W(i)} U_{wt}^i \theta_w^i - p_t^i \geq 0, & (\bar{\lambda}_t^i) & t \in [1, T], i \in \mathcal{G} \\ p_t^i - \sum_{w \in W(i)} L_{wt}^i \theta_w^i \geq 0, & (\underline{\lambda}_t^i) & t \in [1, T], i \in \mathcal{G} \end{cases} \quad (8)$$

$$\begin{cases} \sum_{w \in W(i)} \bar{\Delta}_{wt}^i \theta_w^i \geq p_{t+1}^i - p_t^i, & (\bar{\mu}_t^i) & t \in [1, T-1], i \in \mathcal{G} \\ \sum_{w \in W(i)} \underline{\Delta}_{wt}^i \theta_w^i \geq p_t^i - p_{t+1}^i, & (\underline{\mu}_t^i) & t \in [1, T-1], i \in \mathcal{G} \end{cases} \quad (9)$$

$$\begin{cases} \sum_{i \in \mathcal{G}} p_t^i \geq D_t, & (\alpha_t) & t \in [1, T] \\ \sum_{i \in \mathcal{G}} \sum_{w \in W(i)} U_{wt}^i \theta_w^i \geq D_t + R_t, & (\beta_t) & t \in [1, T] \end{cases} \quad (10)$$

$$- \sum_{w \in W(i)} \theta_w^i \geq -1, \quad (\gamma^i) \quad i \in \mathcal{G} \quad (11)$$

$$p_t^i \in \mathbb{R}_+, \theta_w^i \in \{0, 1\}, \quad t \in [1, T], w \in W(i), i \in \mathcal{G} \quad (12)$$

Extended Formulation of UC

- ▶ Constraints (8), (9), and (10) correspond to constraints (2), (3), and (5), respectively
- ▶ Following **constants** can be easily computed:
 - C_w^i : Start-up cost of unit i under schedule w
 - $U_{w,t}^i/L_{w,t}^i$: Maximum/minimum power output of unit i at time t under schedule w
 - $\overline{\Delta}_{wt}^i/\underline{\Delta}_{wt}^i$ Ramp up/down limit of unit i at time t under schedule w .
- ▶ Constraints (11) restricts that at most **one** schedule can be adopted for each unit

Equivalence between Two Formulations of UC

Theorem 2.1

Let z_{ex}^* and $z_{3\text{-bin}}^*$ be the optimal values of the extended formulation (7)-(12) and the 3-bin formulation (1)-(6), respectively. Then

$$z_{\text{ex}}^* = z_{3\text{-bin}}^*.$$

Moreover, let $z_{\text{ex}}^{\text{LP}}$ and $z_{3\text{-bin}}^{\text{LP}}$ be respective optimal values of their linear relaxations. Then,

$$z_{\text{ex}}^{\text{LP}} \geq z_{3\text{-bin}}^{\text{LP}}.$$

Proof insight: For each unit i ,

- ▶ Establish a **bijection** between spaces $\mathbf{X}^i = \{x^i \in \{0, 1\}^T : (4), (6)\}$ and $\Theta^i = \{\theta^i \in \{0, 1\}^{|W(i)|} : (11)\}$
 - $\mathbf{X}^i \rightarrow \Theta^i$: $\theta_w^i = 1$, if $w = x^i$; $\theta_w^i = 0$, otherwise
 - $\Theta^i \rightarrow \mathbf{X}^i$: $x^i = \sum_{w \in W(i)} \theta_w^i w$
- ▶ Linear relaxation of Θ^i is mapped to $\text{conv}(\mathbf{X}^i)$

Exponential Variables in Extended Formulation of UC

- ▶ The extended formulation is at least as **tight** as the 3-bin formulation (Theorem 2.1)
- ▶ The extended formulation has an **exponential** number of variables (columns)
 - $|W(i)| : \mathcal{O}(2^T)$
- ▶ **Column generation** is commonly used to handle linear programs with many columns

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Column Generation: An Overview

- ▶ **Column Generation** (CG) is a method commonly used in **linear programming** for problems with a large number of decision variables (columns)
- ▶ The key idea: start with a **subset of the columns** and iteratively add more columns to improve the solution
- ▶ Two central components: the **Restricted Master Problem** (RMP) and the **Pricing Problem**

Master Problem (MP)

- ▶ CG is useful when problem has a large number of columns.
- ▶ We want to solve the following large-scale LP problem (MP):

$$\min \sum_{j \in J} c_j \lambda_j \quad (\text{MP.1})$$

$$\text{s. t. } \sum_{j \in J} a_{ij} \lambda_j \geq b_i, \quad \forall i \in I \quad (\text{MP.2})$$

$$\lambda_j \geq 0, \quad \forall j \in J \quad (\text{MP.3})$$

- ▶ $|J|$ very large (or even infinite), directly solving is computationally expensive or infeasible.

Restricted Master Problem (RMP)

- ▶ The RMP is the model with only some columns $J' \subseteq J$
- ▶ Solution to RMP is a feasible solution to the MP

$$\min \sum_{j \in J'} c_j \lambda_j \quad (\text{RMP.1})$$

$$\text{s. t.} \quad \sum_{j \in J'} a_{ij} \lambda_j \geq b_i, \quad [\pi_i] \quad \forall i \in I \quad (\text{RMP.2})$$

$$\lambda_j \geq 0, \quad \forall j \in J' \quad (\text{RMP.3})$$

- ▶ Typically, $|J'|$ small
- ▶ Does λ solve the MP to optimality as well?
 - Sufficient optimality condition: $\bar{c}_j(\pi) = c_j - \sum_{i \in I} \pi_i a_{ij} \geq 0, \forall j \in J$

Adding Columns Iteratively

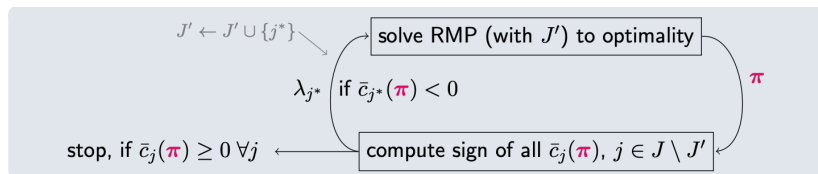


Figure: A natural iterative procedure to solve MP

► How to obtain j^* ?

- Complete enumeration is intractable
- Solve an auxiliary optimization problem, **Pricing Problem**, to determine

Pricing Problem

$$\bar{c}^*(\pi) = \min \{ \bar{c}_j(\pi) \mid j \in J \} \quad (13)$$

- ▶ If $\bar{c}^*(\pi) < 0$, we set $J' \leftarrow J' \cup \arg \min_{j \in J} \{ \bar{c}_j(\pi) \}$ and re-optimize the RMP
- ▶ Otherwise, i.e., $\bar{c}^*(\pi) \geq 0$, there is no $j \in J$ with $\bar{c}_j(\pi) < 0$ and we proved that we solved the MP to optimality

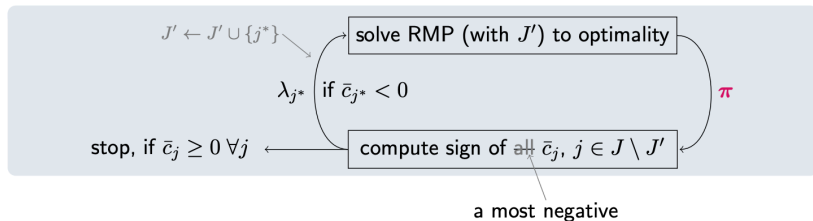


Figure: An iterative procedure with pricing to solve MP

CG Algorithm

algorithm column generation

input: restricted master problem RMP with an initial set $J' \subseteq J$ of variables;

output: optimal solution λ to the master problem MP;

repeat

 solve RMP to optimality, obtain λ and π ;

 solve pricing problem $\bar{c}^*(\pi) = \min\{\bar{c}_j(\pi) \mid j \in J\}$;

if $\bar{c}^*(\pi) < 0$ **then**

$J' \leftarrow J' \cup \{j^*\}$ with $\bar{c}_{j^*}(\pi) = \bar{c}^*(\pi)$; // add variable λ_{j^*} to RMP

until $\bar{c}^*(\pi) \geq 0$;

RMP of Extended Formulation of UC

- ▶ MP: extended formulation (7)-(12) with $\theta_w^i \in \{0, 1\}$ relaxed to $\theta_w^i \in \mathbb{R}_+$
- ▶ $\overline{W}(i)$: partial schedule set for unit i , $\widetilde{W}(i) = W(i) \setminus \overline{W}(i)$
- ▶ RMP: MP obtained by replacing notation $W(i)$ to $\overline{W}(i)$ for $i \in \mathcal{G}$

The dual problem of RMP is

$$\begin{aligned}
 & \max \sum_{t=1}^T D_t \alpha_t + \sum_{t=1}^T (D_t + R_t) \beta_t - \sum_{i \in \mathcal{G}} \gamma^i \\
 & \text{s.t.} \quad \sum_{t=1}^T U_{wt}^i \overline{\lambda}_t^i - \sum_{t=1}^T L_{wt}^i \underline{\lambda}_t^i + \sum_{t=1}^{T-1} \overline{\Delta}_{wt}^i \overline{\mu}_t^i + \sum_{t=1}^{T-1} \underline{\Delta}_{wt}^i \underline{\mu}_t^i + \\
 & \quad \sum_{t=1}^T U_{wt}^i \beta_t - \gamma^i \leq C_w^i, \quad [\theta_w^i] \quad w \in \overline{W}(i), \quad i \in \mathcal{G} \\
 & \quad -\overline{\lambda}_t^i + \underline{\lambda}_t^i + \overline{\mu}_t^i - \overline{\mu}_{t-1}^i - \underline{\mu}_t^i + \underline{\mu}_{t-1}^i + \alpha_t \leq F_t^i, \quad [p_t^i], t \in [1, T], \quad i \in \mathcal{G} \\
 & \quad \overline{\lambda}_t^i, \underline{\lambda}_t^i \geq 0, \quad \overline{\mu}_t^i, \underline{\mu}_t^i \geq 0, \quad \alpha_t \geq 0, \quad \beta_t \geq 0, \quad \gamma^i \geq 0, t \in [1, T], \quad i \in \mathcal{G}
 \end{aligned}$$

Pricing Problem of Extended Formulation of UC

Given values of dual variables , the pricing problem for each $i \in \mathcal{G}$:

$$\min_{w \in \overline{W}(i)} C_w^i - \sum_{t=1}^T U_{wt}^i \bar{\lambda}_t + \sum_{t=1}^T L_{wt}^i \underline{\lambda}_t - \sum_{t=1}^{T-1} \bar{\Delta}_{wt}^i \bar{\mu}_t - \sum_{t=1}^{T-1} \underline{\Delta}_{wt}^i \underline{\mu}_t - \sum_{t=1}^T \bar{U}_{wt}^i \beta_t + \gamma^i \quad (14)$$

By recovering $w \in W(i)$ to x_t^i , s_t^i and z_t^i , (24) can be rewritten as:

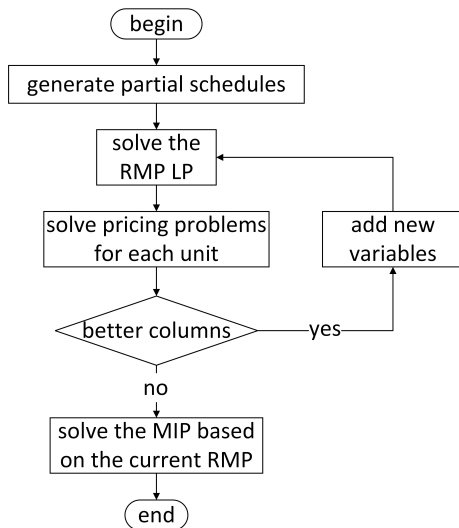
$$\begin{aligned} \min \quad & f(x^i, s^i, z^i) \\ \text{s.t.} \quad & x_t^i - x_{t-1}^i = s_t^i - z_t^i, \quad t \in [2, T] \\ & \sum_{j=t-T_{on}^i+1}^t s_j^i \leq x_t^i, \quad t \in [T_{on}^i + 1, T] \\ & \sum_{j=t-T_{off}^i+1}^t z_j^i \leq 1 - x_t^i, \quad t \in [T_{off}^i + 1, T] \end{aligned} \quad (15)$$

where $f(x^i, s^i, z^i)$ is piece-wisely linear function related to constants in the 3-bin formulation and values of dual variables.

Denote the problem (15) **pricing subproblem** for unit i .

- ▶ Feasible **initial columns** for RMP: $\overline{W}(i) = \{(1, 1, \dots, 1)\}$ for each unit i
- ▶ Pricing subproblems are independent and can be solved in **parallel**.
- ▶ At each iteration, add **all columns** from pricing subproblems, excluding the optimal ones
- ▶ Column generation solves the **linear relaxation** but not the full MIP of extended formulation

Two-stage Column Generation Strategy



Second-stage: Solve MIP based on RMP

How to solve the **MIP**?

- ▶ CG solves the linear relaxation to optimality. Obtain
 - Final RMP
 - LP optimal solution: $\hat{\theta}_w^i, w \in \overline{W}(i), i \in \mathcal{G}$
- ▶ **Restrict Master Integer Problem (RMIP)**: Final RMP with θ_w^i set to be **binary**
 - Optimal solution to RMIP is a **feasible solution** to the extended formulation
- ▶ Large Neighborhood Search Heuristic
 - if $\hat{\theta}_w^i = 0$, set $\theta_w^i = 0$
 - if $\hat{\theta}_w^i > 0.5$, set $\theta_w^i = 1$
 - Solve RMIP with fixing variables

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- ▶ Environment
 - 2.30 GHz Intel(R) Xeon(R) Gold 6140 CPUs and 180 GB RAM
 - Julia 1.7.3, JuMP 0.23.2, CPLEX 20.1.0
- ▶ Configuration
 - Threads: 32
 - MIPGap: 0
 - Time limit: 3600s
- ▶ Test sets: g1000, g1100, g1200, g1300, g1400, g1500 containing 20 instances each (generated cases)
- ▶ Method:
 - 3BIN: directly solving the 3-bin formulation
 - 2SCG: proposed 2-stage column generation method

Overall Performance Comparison

- ▶ Solved: number of instances successfully solved solving within time limit
- ▶ Time: average solving time across all instances in the test set.
- ▶ Obj_Diff: average relative gap of 2SCG. Let z_{3BIN}^* and z_{2SCG}^* be the final solutions found by 3BIN and 2SCG found, respectively. $Obj_Diff = \frac{z_{2SCG}^* - z_{3BIN}^*}{|z_{3BIN}^*|}$

Testset		g1000	g1100	g1200	g1300	g1400	g1500
3BIN	Solved	18	14	13	5	3	2
	Time	1889.66	2450.61	2423.63	3316.40	3334.37	3503.72
2SCG	Solved	20	20	20	20	20	20
	Time	667.26	785.55	890.99	983.64	1102.62	1224.05
	Obj_Diff	-0.011%	-0.012%	-0.010%	-0.010%	-0.011%	-0.011%

Table: Performance Comparison Summary

2SCG can find slightly worse but highly competitive solutions in short time

Statistics of 2SCG

- ▶ CG TPer: Percentage of CG time in total time
- ▶ CG Iter: Number of Iterations in CG
- ▶ NSch: Number of schedule variables in final RMP
- ▶ NFix1: Number of schedule variables fixed to 1

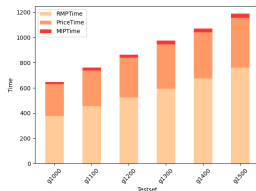


Figure: Time Breakdown for 2SCG

Testset	g1000	g1100	g1200	g1300	g1400	g1500
CG TPer	97.24%	96.96%	97.11%	97.14%	97.18%	97.24%
CG Iter	13.6	13.9	14.05	13.95	14.15	14.1
NSch	8194.4	9066.25	9907.95	10729.9	11573.9	12423.05
NFix1	989.25	1087.45	1185.8	1285.05	1384	1483.3

Table: Statistics of 2SCG

- ▶ CG is **central** to 2SCG's effectiveness in decomposing and solving large- scale problem
- ▶ Despite the exponential scale of $W(i)$, only a **small subset** is added to RMP
- ▶ Most units are assigned a **specific schedule** through our aggressive heuristic approach

Robustness of 2SCG

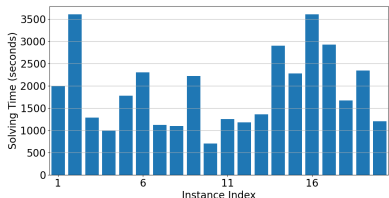
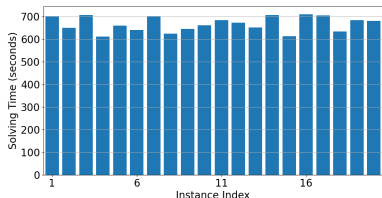


Figure: Comparison of Solving Times for Instances in g1000: 2SCG (left) vs. 3BIN (right)

	g1000	g1100	g1200	g1300	g1400	g1500
3BIN	0.46	0.39	0.42	0.17	0.20	0.10
2SCG	0.05	0.05	0.05	0.05	0.06	0.04

Table: Comparison of Coefficient of Variation on solving time

2SCG shows more robust performance than 3BIN on same-scale instances

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Summary and Future Work

► Summary

- Presented a novel extended formulation of UC and demonstrated its equivalence and tightness
- Proposed a two-stage column generation method for the extended formulation, with numerical experiments highlighting its effectiveness and robustness

► Future Work

- Incorporate a branch-and-price framework to achieve optimality
- Enhance column generation by:
 - Refining the selection of initial columns
 - Accelerating the solution of pricing subproblems (using dynamic programming?)
- Expand the test set for numerical experiments

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Thank you all for your attention!